Increasing electricity exports through delivery-informed allocation in distribution networks

Junjie Huang[1], Tao Yu[1], Zhenning Pan[1], Yufeng Wu[1],
Wencong Xiao[1], Rankai Zhu[1], Jie Huang[2] and Yusheng Xue[3,*]

[1]School of Electrical Engineering, South China University of Technology, and Guangdong Provincial Key Laboratory of Intelligent Measurement and Advanced Metering of Power Grid, Guangzhou 510640, China
[2]NARI Technology Co., Ltd., Nanjing 211106, China
[3]State Grid Electric Power Research Institute (NARI Group Corporation), Nanjing 211106, China
[*]Correspondence: xueyusheng@sgepri.sgcc.com.cn

Abstract

In request-based export sharing, network access is allocated before participants know their available export. An unused award can prevent another participant from exporting, but the shortfall alone cannot establish that loss. We show how a larger request can benefit its sender while reducing total delivery, and propose externality-conditioned counterfactual feedback (ECF) to guide later requests. ECF repeats the allocation after an interval, with unused-award requests reduced to completed delivery. The comparison uses fixed network conditions and a post-event availability record. It tightens future request ceilings only if the alternative is physically feasible and increases delivery elsewhere and in total. In controlled medium-voltage tests across five allocation rules, ECF delivers 24.8 kW more on average than a control that reduces the same total request across all participants. An 80-day simulation combining public Australian network and household data finds 1.39 MWh more modeled delivery than either no feedback or the equal-total control. Applying the same update after every unused award loses 6.42 MWh relative to ECF. At similar frequency, a replay-free control delivers 1.36 MWh less than ECF. Thus the choice of when and where to adjust later requests matters, not just the amount of access withdrawn.

1 Introduction

Distributed generation, including photovoltaic (PV) generation, can supply the wider power system only if local networks can carry the electricity not consumed on-site. Voltage and equipment-loading limits constrain simultaneous export at some locations and times. Operating envelopes translate these constraints into connection-level bounds[1–3], which increasingly form part of flexible export and connection arrangements[4–7]. When several connections share a binding limit, assigning access to one can restrict another. How access is shared then affects not only permission to export, but also how much available electricity reaches the network.

Request-based allocation introduces a timing problem. Participants ask for access before their available export is fully known, and the allocation rule divides access using those requests. A large request protects delivery if output is high, but can leave part of the resulting award unused if output is low. Such a request can reflect uncertainty, private information, or a rational preference to preserve access for a high-output outcome[8–10]. Its value to the participant need not coincide with its effect on other connections.

Fig. 1 shows why an unused award alone cannot reveal lost delivery. Two connections share a binding network constraint. In (a), both use their awards fully. In (b) and (c), connection B leaves the same portion unused, but the consequences differ. Giving A more access would not increase delivery in (b),

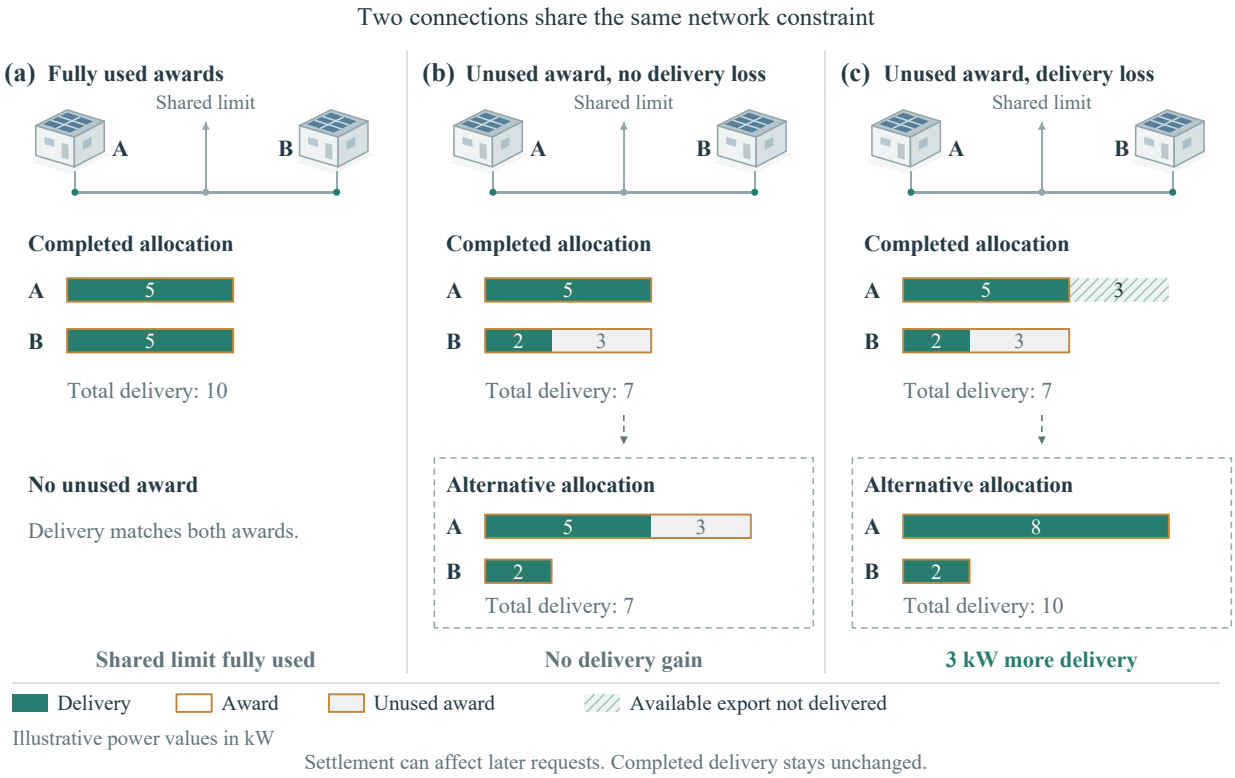


Figure 1: When unused export permission displaces delivery. Two connections share a binding network constraint. (a) Both awards are fully used. (b) B leaves part of its award unused, but A has no additional available export. An alternative allocation increases A's award without increasing delivery. (c) B leaves the same portion unused while A has available export that its award prevents it from delivering. The alternative allocation increases total delivery. Teal fills show delivery, amber outlines show awards, and hatching shows available export not delivered. Numbers within each segment give its power in kW. Dashed frames show calculations for the completed interval. Settlement can affect later requests but cannot restore that interval's lost delivery. The illustrative values use a shared 10 kW authorization bound and initial awards of 5 kW each; they are not case-study results.

because A has no additional electricity available. In (c), A could use that access, so an alternative allocation under the same conditions would allow more delivery. An award and its meter reading reveal the shortfall, but cannot distinguish these two situations.

Neither charging for a shortfall nor reducing all requests answers this allocation question on its own. A shortfall charge changes the participant's incentive to request access, whereas the delivery lost elsewhere depends on who could have used it. Broad reductions can avoid unused awards but also remove access needed during high-output intervals. The operational challenge is therefore to use the information revealed after delivery to improve later allocation, while retaining useful access under uncertainty.

Measurements can guide subsequent power-system actions[11,12]. In export sharing, metered delivery identifies unused awards, while a record of available export helps determine whether another connection could have used the access. A later request update must still allow for output to recover.

Operating-envelope methods turn voltage and equipment limits into export bounds, including bounds

that account for uncertain feeder conditions[1,2,13,14]. Regulatory guidance and demonstration specifications describe how flexible limits can be issued and exchanged[15–17]. Within those limits, allocation rules share access using proportional, equitable, or bargaining objectives[18–21]. The choice of objective affects who receives an award[22–25], but network feasibility does not tell the operator how much of that award will be used once output is realized.

Other approaches act at different points in the operating cycle. Network-checked exchange can transfer an issued limit before delivery[26]. Local measurements, past curtailment and persistent allocation states can guide later limits or objectives[27–29]. These methods use valuable operating information, but the information serves different decisions. ECF uses a completed interval to test whether changing its allocation would have increased delivery elsewhere; that result determines whether to lower a later request ceiling. This comparison requires an estimate of available export under the alternative allocation.

Requests also respond to incentives. Energy-sharing and flexibility studies link network constraints with bids, prices, and uncertain delivery[8,30–35]. Work on reported baselines likewise shows that private incentives can change what participants report[9,10,36]. A shortfall charge can change requests before allocation. It cannot, from the participant's shortfall alone, measure electricity forgone at other network connections. That is the comparison needed to guide a later request limit.

We first identify when protecting a participant's own export reduces total delivery. A single uncertain-output model generates both request choice and later delivery, linking the incentive for a larger request to its effect on other connections. This request–access–delivery externality distinguishes productive larger requests from those that increase the requesting group's payoff and unused award while lowering total delivery. The analysis gives conditions for this divergence and evaluates finite changes when active network constraints change.

We then introduce externality-conditioned counterfactual feedback (ECF), which uses recoverable delivery rather than shortfall alone to select later request updates. It compares the completed allocation with a physically feasible alternative and tightens future ceilings only when that alternative recovers delivery elsewhere and increases the total. The update blends the current ceiling with a target derived from the completed interval, while leaving the allocation rule unchanged. We derive the ceiling bounds and recovery rate. The comparison identifies recoverable delivery for the unused-award group as a whole; it neither assigns individual responsibility nor requires private payoff information.

Three comparisons probe why feedback changes delivery. An equal-total control spreads the same request reduction across participants. A second control uses ECF's update after every unused award. A third applies that update at a similar frequency, but chooses its intervals without replay. Together they examine the location, frequency and information used in later request reductions. Methods defines these comparisons; the Supplementary Information gives proofs and numerical details.

Fig. 2 places ECF in the operating sequence. Requests are allocated before available power is known; delivery then reveals unused awards. Neither a shortfall nor a later penalty shows whether another connection lost usable access. ECF therefore repeats the allocation with the unused-award group's requests reduced to completed delivery. It holds the exogenous network conditions and allocation rule fixed, while a post-event available-export estimate tests delivery elsewhere and in total. The estimate must remain valid under the alternative award. A qualifying result lowers the group's next request ceiling; otherwise ceilings recover. The completed award never changes.

The controlled study supplies availability from simulation. The Australian study constructs it from public PV and load profiles, so delivery there is modeled rather than metered at the assessed connections. The replay test is distinct from the private-payoff classification of larger requests.

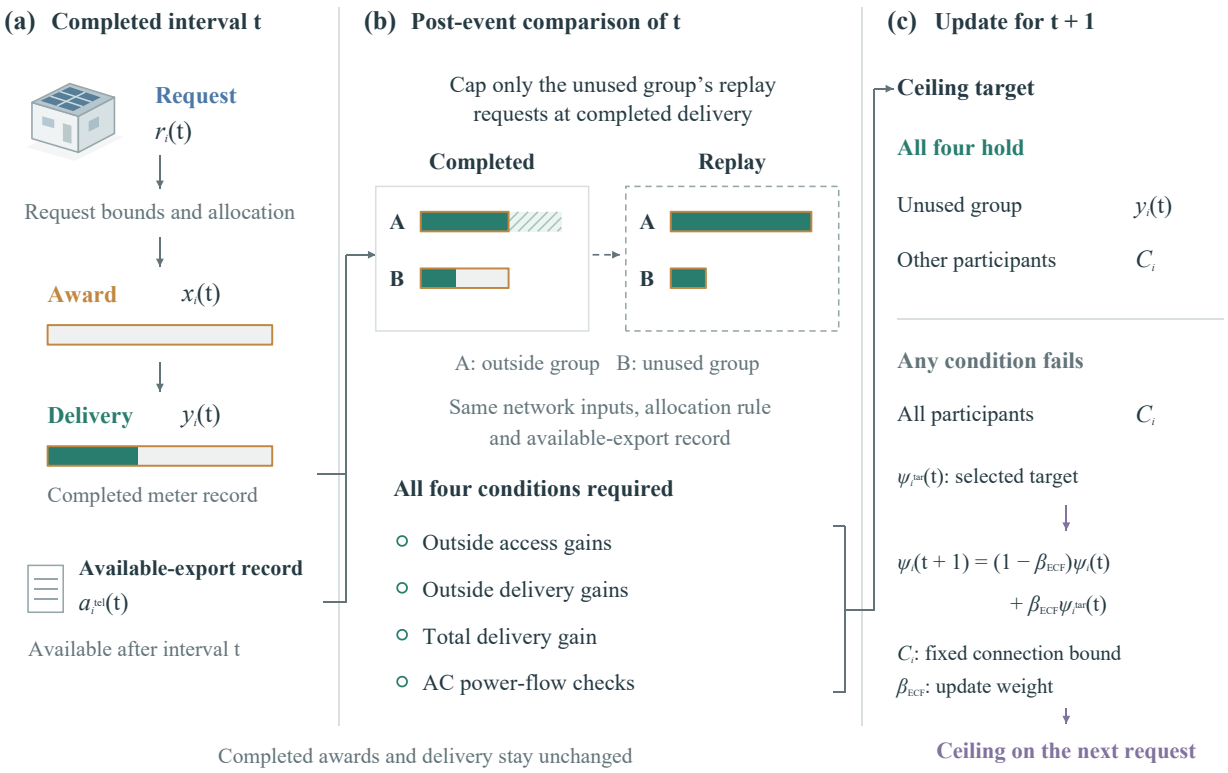


Figure 2: Using completed delivery to update later request ceilings. (a) Submitted requests pass through connection and current feedback bounds before the existing rule allocates access. (b) After delivery, requests from the unused-award group are reduced to completed delivery and allocation is repeated under the same exogenous network conditions. The comparison uses a post-event available-export record and checks whether delivery could rise outside the group and in total without violating the specified AC limits. (c) A qualifying comparison sets a lower target for the group's next request ceiling; otherwise, ceilings recover toward connection bounds. The completed award and delivery do not change.

Table 1 distinguishes the comparison mechanisms. The equal-total control tests where requests are reduced. The unqualified and frequency-calibrated random updates test what is gained by selecting update intervals through replay. Other controls change request incentives, use a participant's delivery ratio, or omit the forecast increment.

2 Results

The experiments address two decisions: where a reduction in requested access should fall, and when a completed shortfall warrants a later reduction. The controlled study tests request choice and five allocation rules. An Australian public-data simulation then compares ECF with no feedback, an equal-total reduction, and updates made without the replay test. We report mean power in kW for controlled intervals and cumulative modeled energy in MWh for the Australian assessment. Each difference names its comparator; all reported differences are calculated before rounding.

Table 1: How the comparison mechanisms differ. C denotes the controlled study and A the Australian simulation. Each paired comparison uses the same network and available-export trajectory.

Mechanism	What changes	Question tested
No feedback (C, A)	Ceiling stays at the connection bound.	Baseline delivery.
Shortfall cost (C)	A modeled cost changes the initial request.	Effect of an incentive before allocation.
Mismatch-only feedback (C)	A participant's own delivery-to-award ratio updates its ceiling.	Effect of responding to shortfall alone.
Update without replay test (A)	ECF's formula applies after every unused award.	Effect of removing qualification, including more frequent updates.
Frequency-calibrated random update (A)	The same formula updates randomly selected unused-award intervals.	Whether a similar update frequency explains ECF's gain.
Equal-total reduction (C, A)	ECF's total reduction is spread proportionally.	Value of where requests are reduced.
Previous-day persistence (A)	Request uses the previous-day output estimate alone.	Cost of a conservative request.
ECF (C, A)	Qualified unused awards lower later ceilings.	Combined effect of testing and updating.

2.1 Private delivery gains can accompany a larger loss elsewhere

Larger requests can benefit the requesting participant while reducing total delivery. Across 1,500 controlled parameter–location cases without a shortfall cost, the participant gains 5.2 kW of expected delivery on average while others lose 18.8 kW. Total delivery falls by 13.5 kW. The request model values the participant's own gain but does not include the larger loss elsewhere. Because the same uncertain-output model determines both request choice and expected delivery, this difference arises from the allocation response rather than from unrelated request and output curves.

Fig. 3 shows how this response changes with forecast uncertainty σ and normalized shortfall cost ϑ. The latter is the modeled cost per unit of unused award divided by the value per unit of delivered export. We compare weighted proportional (WP), equal reduction (ER), equal access (EA), maximum export (ME), and fairness-aware (FA) allocation under two reactive-power settings. Each selected request is paired with the same participant's forecast-based reference. Methods gives the five-outcome available-export model and exact finite-difference calculation.

The aggregate loss does not make every larger request unproductive. We classify a request–access–delivery externality when expected payoff and unused award both rise while total delivery falls. A productive high request instead raises payoff without lowering total delivery beyond the numerical tolerance. For $\vartheta \geq 0$, the externality conditions imply that the requesting participant delivers more and others lose still more; Methods and Section S2 of the Supplementary Information prove this implication and specify the classification.

Both high-request classes occur in Fig. 3. Among the 1,500 controlled parameter–location cases at $\vartheta = 0$, pooled over all five uncertainty levels, 71.5% meet the externality definition and 11.9% are productive high requests. Reference requests account for 14.7%, and the remaining 1.9% are larger requests that meet neither high-request definition. Section S2 reports the corresponding counts. The coexistence of these classes explains why request size alone is insufficient to identify a delivery loss.

Uncertainty and the normalized shortfall cost shift the relative frequency of these outcomes in

opposite directions. With greater forecast uncertainty, a large award can accommodate a higher-output realization. Greater ϑ makes unused awards more costly, so the selected request falls. At $\vartheta = 10$, 10.7% of the 1,500 cases meet the externality definition. Across all cases, the mean total-delivery loss, counting cases without a loss as zero, falls to 0.6 kW. Productive high requests still account for 11.5% of the cases. A shortfall cost can therefore reduce requests that lower total delivery while preserving some productive requests above the forecast.

Panel (d) asks how much delivery is lost elsewhere per extra unit of unused award. For each 0.1 increase in the request multiplier, it divides the first quantity by the second. The ratio $\vartheta^{\text{ext}}_{i,\Delta}$ is the normalized shortfall cost that would equate the private-payoff change with the total-delivery change for that request step. Supplementary Section S2 derives it and specifies when its denominator is positive. The calculation covers 4,608 steps in which own delivery and unused award rise while total delivery falls. The median ratio is 2.29, with an interquartile range of 1.10–5.72. Rule-specific medians range from 0.92 for ME to 5.37 for EA. This spread shows that one system-wide shortfall cost cannot match the cross-participant delivery effect across all tested electrical locations and operating states. It motivates the post-event feedback comparison in the next section.

2.2 Targeted request reductions preserve useful export access

At $\beta_{\text{ECF}} = 0.5$ and $\vartheta = 0$, ECF delivers 24.8 kW more on average than the equal-total control. Thus its advantage over uniform reduction cannot be explained by removing more total request. The shortfall-cost and mismatch-only comparisons ask whether changing request incentives or reacting to each participant's own delivery provides the same benefit. Controls begin with identical raw requests and allocations. Feedback can affect only later intervals; the unused-award group is identified anew after each one.

The update cannot affect the allocation that produced its evidence. A representative trajectory in Supplementary Fig. 1 follows five nearby participants through a drop and recovery in available export. Delivery recovered by the completed-interval comparison and delivery in later intervals are separate outcomes. The three controlled temporal profiles use the same available-export values with different orders, so their comparison tests timing rather than a change in the output distribution.

2.2.1 A shortfall cost reduces displacement but also useful delivery

A shortfall cost reduces requests associated with displacement but also removes useful access (Supplementary Fig. 2). Raising ϑ from 0 to 1 lowers the mean request multiplier from 1.51 to 1.14 and the share of intervals meeting the economic externality definition from 50.1% to 15.6%. It also removes 45.6% of the changed-request group's delivery gain. At $\vartheta = 10$, no interval meets that definition, but the group's gain is almost zero. The externality share is an economic classification, not ECF's activation rate.

At $\vartheta = 0$, the shortfall-cost benchmark equals no feedback. Relative to that common baseline, mismatch-only feedback increases mean total delivery by 15.2 kW and ECF by 24.9 kW. ECF therefore delivers 9.7 kW more. Against forecast-based reference requests, it turns a 21.7-kW no-feedback delivery loss into a 3.2-kW gain while retaining 79.4% of the changed-request group's no-feedback delivery gain. Supplementary Section S3 gives the group decomposition and results at the other shortfall-cost levels.

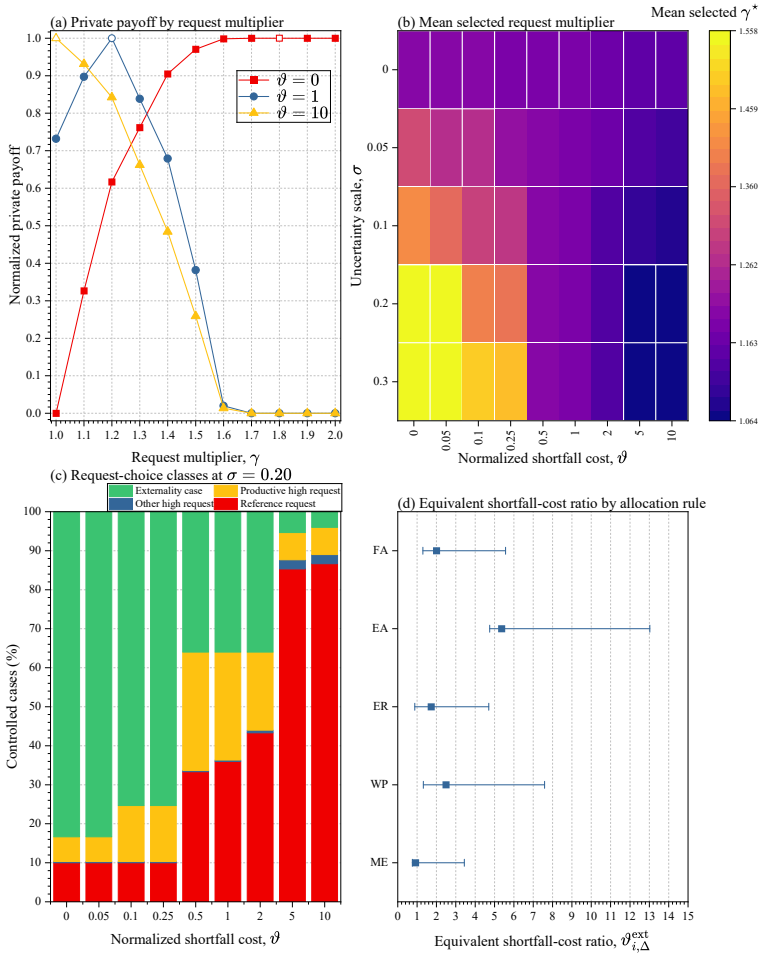


Figure 3: How a larger request becomes productive or harmful before feedback. (a) Representative private-payoff curves for three normalized shortfall-cost levels. Within each curve, payoff is rescaled as $(U - U_{\min})/(U_{\max} - U_{\min})$; a zero span is plotted as zero. (b) Mean selected request multiplier for each uncertainty and shortfall-cost setting. The mean covers participants, allocation rules, and reactive-power settings. (c) Shares at $\sigma = 0.20$ ($n = 300$): reference requests; productive high requests, which raise payoff without lowering total delivery; request–access–delivery externality cases, which raise payoff and unused awards while lowering total delivery; and other high requests, which meet neither high-request definition. (d) Finite-difference cost ratio $\vartheta^{\text{ext}}_{i,\Delta}$: expected delivery loss elsewhere per unit increase in expected unused award for a 0.1 request-multiplier step. Squares show rule-specific medians, and bars show interquartile ranges. WP, ER, EA, ME, and FA denote weighted proportional, equal reduction, equal access, maximum export, and fairness-aware allocation, respectively.

2.2.2 Where requests are reduced determines recovered delivery

ECF and the equal-total control remove the same total request in every interval but reduce different participants' requests. The comparison holds the allocation rule, network conditions and available output fixed. It therefore isolates the delivery effect of where requests are reduced. Mismatch-only feedback instead tests an update based on own delivery without checking recovery elsewhere.

At $\vartheta = 0$ and $\beta_{\text{ECF}} = 0.5$, ECF delivers 9.7 kW more than mismatch-only feedback and 24.8 kW

more than the equal-total control. ECF delivers more in 73.3% and 95.6% of the matched trajectories ($n = 90$ per comparison), respectively. Its mean advantage over both controls remains positive at $\vartheta = 1$ and 10.

A favorable replay does not guarantee a favorable later update. The feedback gain controls how much weight the latest interval receives. Delivery changes nonmonotonically with that weight. At $\vartheta = 0$, the mean change relative to forecast-based requests is -2.8 kW for $\beta_{\mathrm{ECF}} = 0.25$. It rises to 3.2 kW at 0.5 but falls to -23.4 kW at 1. The same reversal occurs at $\vartheta = 1$ and 10. The replay evaluates a completed output realization, whereas the resulting ceiling applies to a different interval. At unit gain, the previous ceiling is discarded in favor of that completed interval's target. The negative response establishes the importance of this temporal distinction; it does not by itself identify a single cause of the reversal. The intermediate weight 0.5 is the primary comparison setting. It gives the largest mean delivery among the three tested weights in all three shortfall-cost settings, but is not a universal optimum.

Of 12,960 controlled replay attempts using exact simulated available export, 12,910 meet all four conditions. Their median total-delivery recovery is 37.8 kW, but an unused award alone does not trigger an update: at the primary gain, 20 of 4,320 attempts fail because total delivery does not rise. The Australian replay qualifies 12.6% of assessed intervals. These rates describe the respective experimental conditions, not a universal activation frequency.

Fig. 4 reports the gain sweep, paired control comparisons, and replay-recovery distribution behind these main comparisons.

2.3 Output timing and allocation rules alter the delivery gain

At $\vartheta = 0$ and $\beta_{\mathrm{ECF}} = 0.5$, the mean ECF delivery advantage is positive at every factor level reported in Fig. 5. This holds for every allocation rule, temporal profile, spatial case, and reactive-power setting. Against no feedback, the rule-specific gain ranges from 8.9 kW for FA to 67.5 kW for EA. Against the equal-total control, it ranges from 9.7 to 50.8 kW. The positive direction is therefore not confined to one allocation rule.

The three temporal profiles contain the same available-export values and frequencies. Only their order changes. The mean delivery advantages over no feedback are 35.3 kW for the clustered profile, 12.2 kW for the intermittent profile, and 27.3 kW for shock–recovery. Against the equal-total control, the corresponding advantages are 34.4, 12.6, and 27.4 kW. Because each profile preserves the available-export values and their frequencies, these differences isolate temporal ordering rather than a change in the uncertainty distribution. A completed low-output interval has a different value for later access when similar states are clustered than when they are separated by other output states.

The mean delivery advantage over no feedback is 11.2 kW when one participant changes its request. It is 35.0 kW for five electrically local participants and 28.6 kW for five dispersed participants. Every participant still selects its own request multiplier. The local–dispersed comparison holds group size fixed but changes participant identities, electrical locations, and participant-specific multipliers. The difference therefore reflects the selected participants as well as their electrical placement. At the reported precision, zero reactive power (Q0) and 0.95-power-factor absorption (Q95) both give a 24.9-kW mean advantage over no feedback. Both also give a 24.8-kW mean advantage over the equal-total control. The two comparator-specific results are therefore unchanged across the two tested reactive-power settings.

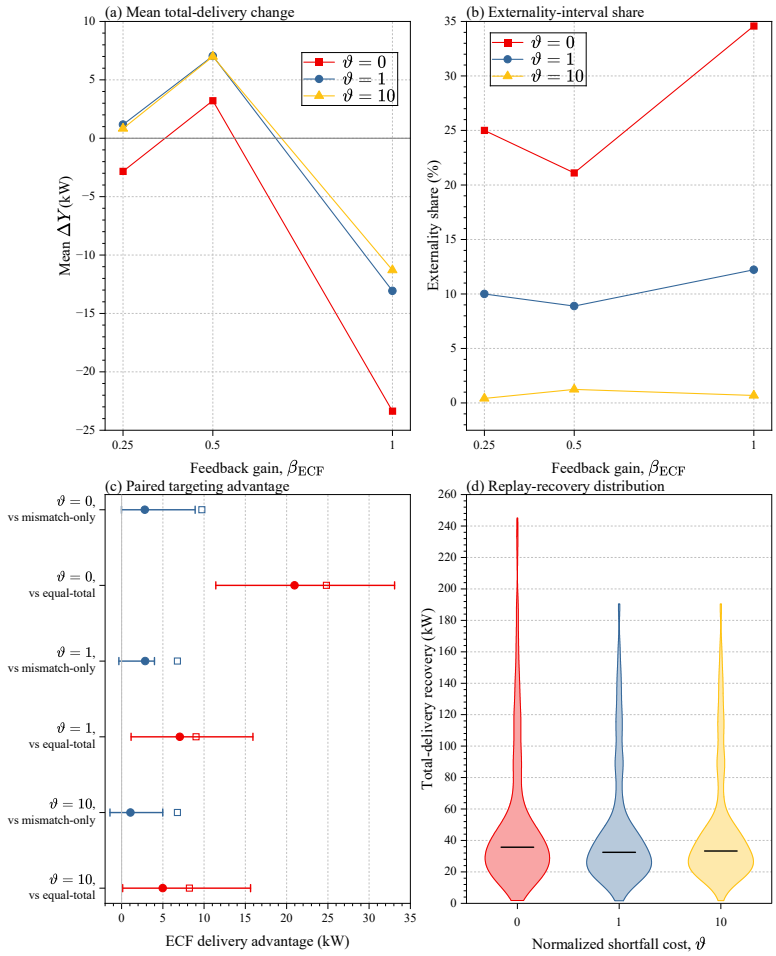


Figure 4: Feedback-gain response, paired controls, and replay recovery across all controlled dynamic cases. (a) Mean signed change in total delivery and (b) externality-interval share for the three tested feedback gains. The series denote $\vartheta = 0, 1, 10$. (c) ECF delivery advantage in matched trajectories. Rows labelled "vs mismatch-only" compare ECF with mismatch-only feedback. Rows labelled "vs equal-total" compare ECF with the equal-total control. Panel (c) uses the primary gain, $\beta_{\text{ECF}} = 0.5$. Filled circles show medians, open squares show means, and whiskers show interquartile ranges. The gray line marks zero. (d) Distribution of total delivery recovered when the replay meets all four conditions. Violin width shows empirical density, and the black horizontal segment shows the median.

2.3.1 AC feasibility and allocation equality

The delivery comparisons remain within the declared physical envelope. Both sides of all 25,920 matched allocation comparisons pass nonlinear AC checks for convergence, active- and reactive-power balance, voltage, and two-ended branch apparent power. Since delivery can be below the award, we separately check the resulting connection injections. All 22,520 distinct observed- and replay-delivery states pass the same checks, with maximum branch utilization of 90.91% and no excess above the fixed synthetic ratings. Section S3.5 of the Supplementary Information gives the state counts and voltage extrema. Supplementary Fig. 3(a) shows the voltage margin: the minimum bus-voltage magnitude minus the 0.95-per-unit (pu) lower limit. Physical feasibility thus holds for

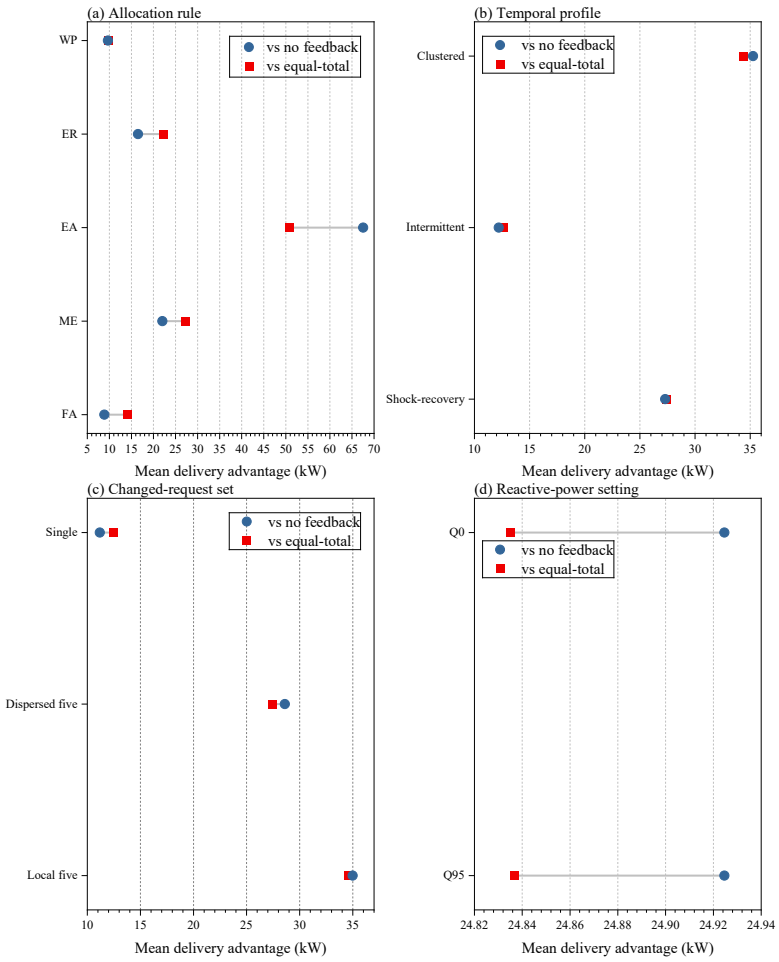


Figure 5: Mean ECF delivery advantage at $\vartheta = 0$ and $\beta_{\mathrm{ECF}} = 0.5$. Results are grouped by (a) allocation rule, (b) temporal profile, (c) changed-request set, and (d) reactive-power setting. Blue circles compare ECF with no feedback. Red squares compare ECF with the equal-total control. WP, ER, EA, ME, and FA denote weighted proportional, equal reduction, equal access, maximum export, and fairness-aware allocation, respectively. Q0 sets participant reactive power to zero, whereas Q95 uses 0.95-power-factor absorption.

both authorized and delivered export, including cases where feedback reduces delivery.

More delivery does not automatically mean more equal access. Supplementary Fig. 3 therefore reports both outcomes. The Jain index measures capacity-normalized allocation equality, not broader social fairness. At $\vartheta = 0$, ECF yields a higher mean Jain-index change than no feedback under all five rules, but most changes remain negative relative to the forecast-based reference allocation. For EA, the change moves from -2.4 percentage points without feedback to 0.1 percentage points with ECF. The ECF values for WP, ER, ME, and FA range approximately from -0.8 to -0.5 percentage points: ECF reduces the decline rather than producing an absolute increase under these four rules. ECF also exceeds the equal-total control on this mean metric for every rule. Panel (c) shows that mean delivery and allocation-equality outcomes vary across rule–control combinations. Delivery and allocation equality can therefore move in different directions. The Supplementary Information reports the corresponding unused-award, allocation-movement, and total-authorization results.

2.4 Public-data replay gains delivery through a different allocation of access

2.4.1 Better use of access increases delivery, not just utilization

ECF also outperforms equal-total reduction when requests and available output follow public household profiles. Across 80 assessed network-days, it delivers 239.36 MWh of modeled export, 1.39 MWh (0.59%) more than each of no feedback and the equal-total control (Fig. 6). The simulation combines a published Australian MV/LV network with measured PV and load chronology. Available export and requests are constructed from these profiles, rather than taken from linked operating records. Allocation uses a common proportional scale, with daily ceiling initialization in the primary comparison. Methods specifies the construction; the following comparisons examine what changes the delivery gain.

More delivery here accompanies more authorization, rather than a smaller total award. Relative to no feedback, authorization increases by 1.206 MWh and unused authorization falls by 0.186 MWh. Together these changes account for the 1.392-MWh delivery gain in (8). Selective request reduction therefore improves delivery without requiring a reduction in total authorization. Supplementary Section S11.1 gives the complete energy balance, including the almost identical contrast with the equal-total control.

Increased authorization is the larger term in this energy balance. Equal admitted-request totals do not fix the total award: different request distributions can admit different proportional allocation factors. Conversely, uniformly scaling requests preserves their relative sizes. If the network constraint still binds, the allocation factor can compensate for that scaling and leave awards unchanged. The near-equality of no-feedback and equal-total delivery is consistent with this property, derived in Supplementary Section S4. Selective updates change relative requests instead. The energy balance records the combined outcome; it does not assign separate causal effects to increased authorization and reduced unused awards.

The gain is uneven over time. ECF delivers more on 44 days, less on 32, and the same amount on four against each control. Its mean daily gain is 17.4 kWh, with dependence-adjusted 95% intervals of 5.1–29.7 kWh against no feedback and 4.9–29.9 kWh against equal-total reduction. Only 483 of 3,840 half-hours (12.6%) qualify for tightening. This frequency describes how often the model finds a recoverable opportunity; the daily comparisons measure the subsequent delivery outcome. The longest contiguous 35-day block contributes 1.06 MWh of the 1.39-MWh gain. The other 45 days contribute 0.33 MWh; their mean-difference interval includes zero when analyzed alone (Supplementary Section S10.3).

The connection-point gain also appears at the source terminal. Net source exchange increases by 1.377 MWh against each control, retaining about 98.9% of the respective delivery gains. The feeder remains a net importer, so this change means reduced upstream import. It includes the small increase in circuit losses and changes in voltage-dependent passive-element exchange; Supplementary Section S11.1 separates those quantities. The delivery gain is therefore visible at the feeder boundary, not only in the connection-level allocation totals.

Authorization utilization, E_Y/E_X, is 72.6% under ECF and 72.4% under each of no feedback and the equal-total control. Here E_Y and E_X are delivered and authorized export energy over the assessed days. Previous-day persistence gives a useful contrast: it omits the forecast increment and uses 89.0% of its authorization, yet delivers 50.33 MWh less than ECF. A high utilization ratio alone can therefore conceal delivery lost through an overly restrictive request. This contrast makes utilization insufficient as an operating objective: withdrawing access can improve the ratio even while reducing delivered energy.

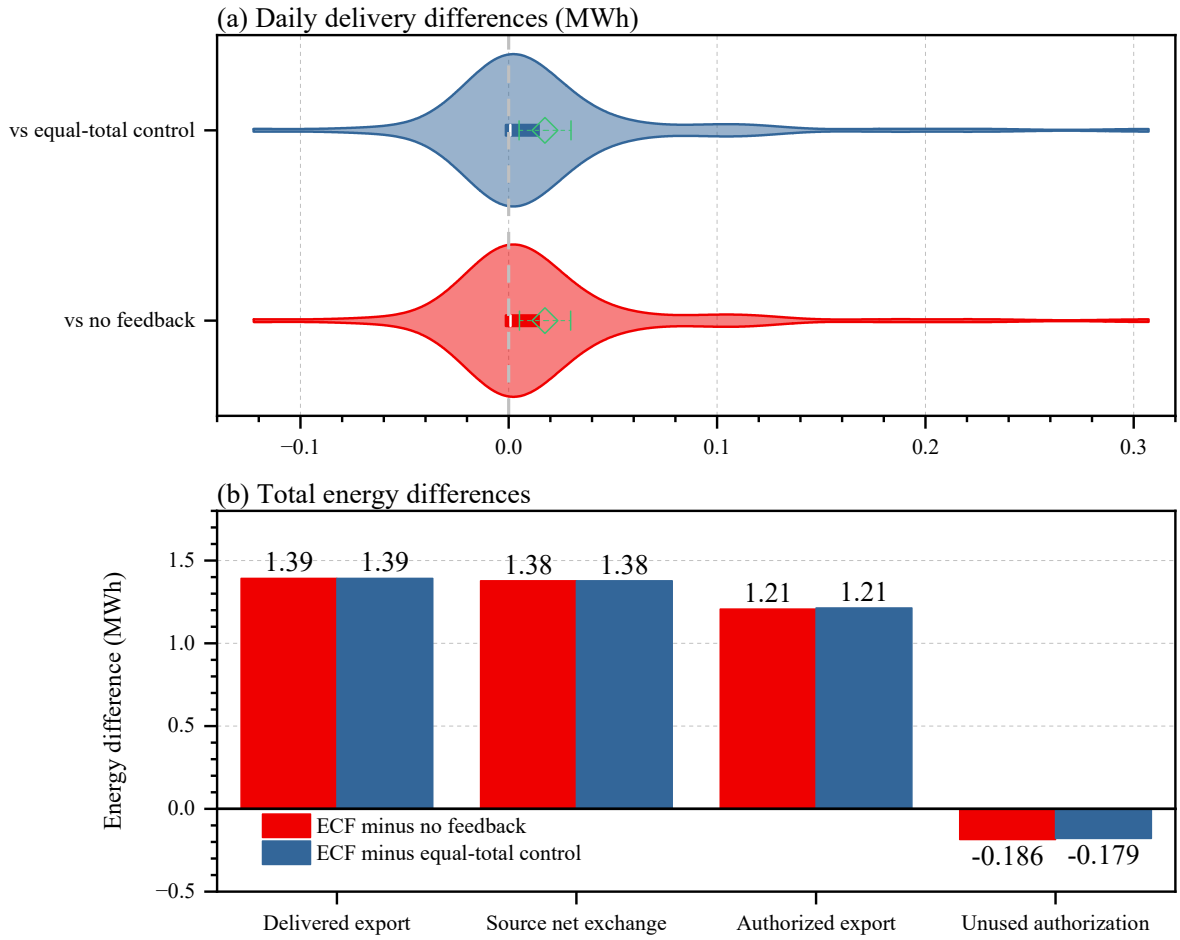


Figure 6: Modeled delivery and authorization changes in the Australian public-data replay. Available export is constructed from PV and load profiles. (a) Paired daily delivery differences for ECF against no feedback and the equal-total control. Violin width shows empirical density. Dark boxes span the 25th–75th percentiles, and white ticks show medians. Green diamonds show means; capped green lines show 95% intervals adjusted for short-range dependence between neighboring dates. (b) Aggregate 80-day differences in delivered export, source-terminal net exchange, authorization, and unused authorization. Each difference is ECF minus the indicated control. Delivery equals authorization minus its unused portion. Here the delivery gain accompanies increased authorization and reduced unused authorization. Positive source-terminal differences indicate reduced net upstream import over the assessed period.

All primary control trajectories satisfy the specified allocation and delivery checks, while network limits reduce the admitted-request scale in 44.4% of ECF half-hours. Thus the common operating limits still constrain allocation. Fig. 6 reports the daily differences and aggregate energy changes.

2.4.2 Replay qualification avoids overrestricting later requests

Keeping the update formula fixed allows two tests of replay qualification. Updating after every unused award, without a replay test, reduces modeled delivery from 239.36 to 232.93 MWh. ECF delivers 6.42 MWh (2.76%) more, with higher delivery on 75 of 80 days. That contrast also changes

Table 2: Modeled export over the same 80 Australian network days. The unqualified update uses ECF's ceiling formula and gain after every unused award. The frequency-calibrated random row averages three replay-free schedules with the same formula and gain. The equal-total control matches ECF's request reduction in each half-hour. All energy totals are in MWh and are rounded independently.

Control	Authorized	Unused	Delivered
No feedback	328.490	90.526	237.963
Equal-total reduction	328.483	90.519	237.964
Update without replay test	305.159	72.226	232.933
Frequency-calibrated random update	328.314	90.318	237.996
ECF	329.696	90.341	239.356

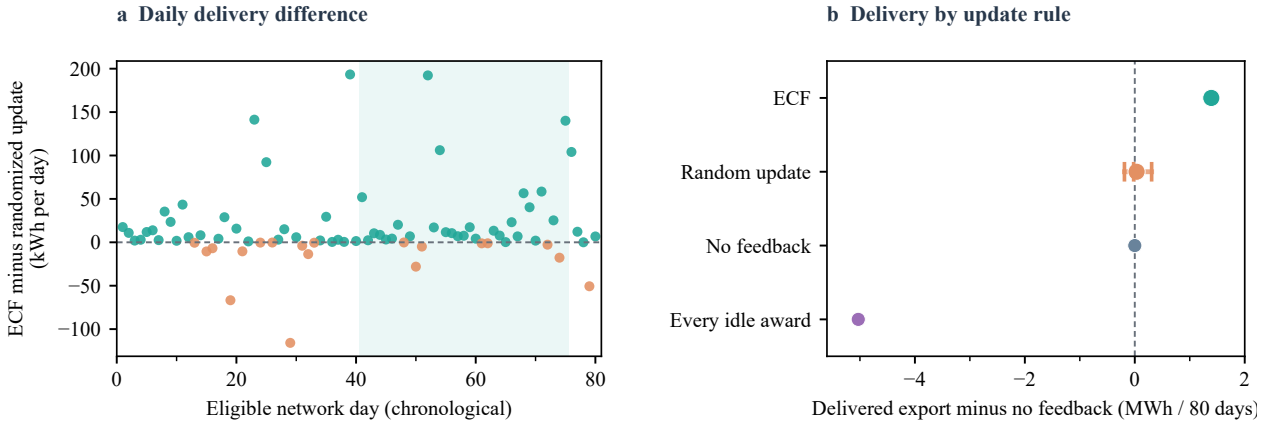


Figure 7: Replay qualification compared with replay-free updates in the Australian public-data simulation. (a) Paired daily modeled-delivery difference between ECF and the mean of three frequency-calibrated random updates. Positive points favor ECF; shading marks the longest continuous 35-day block. (b) Eighty-day delivery relative to no feedback. The central random marker is the mean across three fixed schedules; its horizontal range shows their individual totals. The number of updated half-hours is 483 for ECF, 484–495 for the random schedules, 0 for no feedback and 3,086 for updating after every unused award. The randomized control was designed after the original result and tests update selection, not a new blind cohort.

how often ceilings are tightened: 483 times under ECF, compared with 3,086 times without qualification. The unqualified update leaves 18.11 MWh less authorization unused, but grants 24.54 MWh less authorization. It removes useful delivery along with idle awards.

The frequency-calibrated random control addresses this difference. It applies ECF's target and gain after randomly selected unused awards, without testing recovered delivery. The three fixed schedules update in 484–495 half-hours, close to ECF's 483. Their mean adds just 0.03 MWh over no feedback; ECF adds 1.39 MWh. ECF exceeds the random mean by 1.36 MWh and each schedule individually. After averaging schedules within each date, ECF delivers more on 61 of 80 days; the date-dependent 95% interval for the mean difference is 7.6–26.4 kWh per day. This supports selecting intervals by replay beyond merely updating less often within the tested loop, but does not isolate each replay condition. Full comparisons are in Supplementary Sections S11.2–S11.3.

The replay check determines whether to tighten ceilings; the gain sets the update size. At $\beta_{\mathrm{ECF}} =$

0.25, ECF gains 1.34 MWh over each of no feedback and equal-total reduction. At unit gain, it instead loses 3.04 MWh relative to no feedback. Full replacement of the ceiling by the latest target can restrict export when output recovers. The replay must therefore identify a genuine missed opportunity, and the subsequent update must preserve access for changing output.

The gain reversal persists when ceilings carry across midnight. In ten consecutive-date blocks totaling 80 days, the longest spanning 35 days, total delivery at each of the three gains differs from daily initialization by less than 0.1 kWh. Daily initialization does not explain either the primary advantage or the unit-gain loss (Supplementary Section S12.2).

ECF's mean advantage persists when participants revise requests from their own delivery history. Across 80 dates, it delivers 1.44 MWh more than no feedback and 4.53 MWh more than the same update without qualification. The controls develop separate request histories; this comparison does not establish that adaptation increases the advantage over no feedback. The request model and uncertainty are reported in Supplementary Section S14.2.

2.4.3 Record error changes which intervals qualify

Scaling the post-event record by ±10%, delaying it by one interval, or substituting modeled delivery for 10% of records leaves a positive mean delivery comparison against both controls (Fig. 8). Mean gains span 16.3–18.0 kWh per network day, with dependence-adjusted 95% intervals above zero. Individual replay decisions have different margins to record error, however; a robust mean does not make every update reliable.

A conservative lower bound avoids false recovery qualification in the tested overestimation cases, but can miss real opportunities under underestimation. At the largest tested error radius, it misses 258 of 496 feasible recoveries and delivers 269 kWh less than the point-record control. Both treatments still improve total delivery over no feedback, and their assessed states pass the specified physical checks (Supplementary Section S12.3). Protecting the replay decision and maximizing subsequent delivery are different objectives.

Fig. 8 collects the gain and record responses, together with the complete-customer cohort. The latter broadens the calendar to 361 days at the same load multiplier of 0.30. ECF gains 10.96 MWh over each of no feedback and equal-total reduction, or 30.4 kWh per network day; both mean-difference intervals remain above zero and the specified physical checks pass. Customer composition and calendar coverage change together, so this comparison does not isolate either factor (Supplementary Section S10.4).

2.4.4 Spatial placement and available generation shape the benefit

The delivery advantage persists when the same household profiles occupy different service connections. Across five alternative mappings, ECF gains about 0.9–5.1 MWh over either control during the same 80 days; all pointwise mean-difference intervals are positive. Placement changes the size of the gain, not its sign in this five-mapping comparison. In the least favorable mapping, ECF leaves 0.35 MWh more authorization unused yet delivers 0.86 MWh more than no feedback. Unused authorization alone again fails to predict delivery (Supplementary Section S13.1).

The advantage also persists across PV capacities. At five levels spanning half to one-and-a-half times the primary capacity, ECF gains 1.39–1.88 MWh over either no feedback or the equal-total control during the same 80 dates. The gain is not monotonic in installed capacity and falls per installed

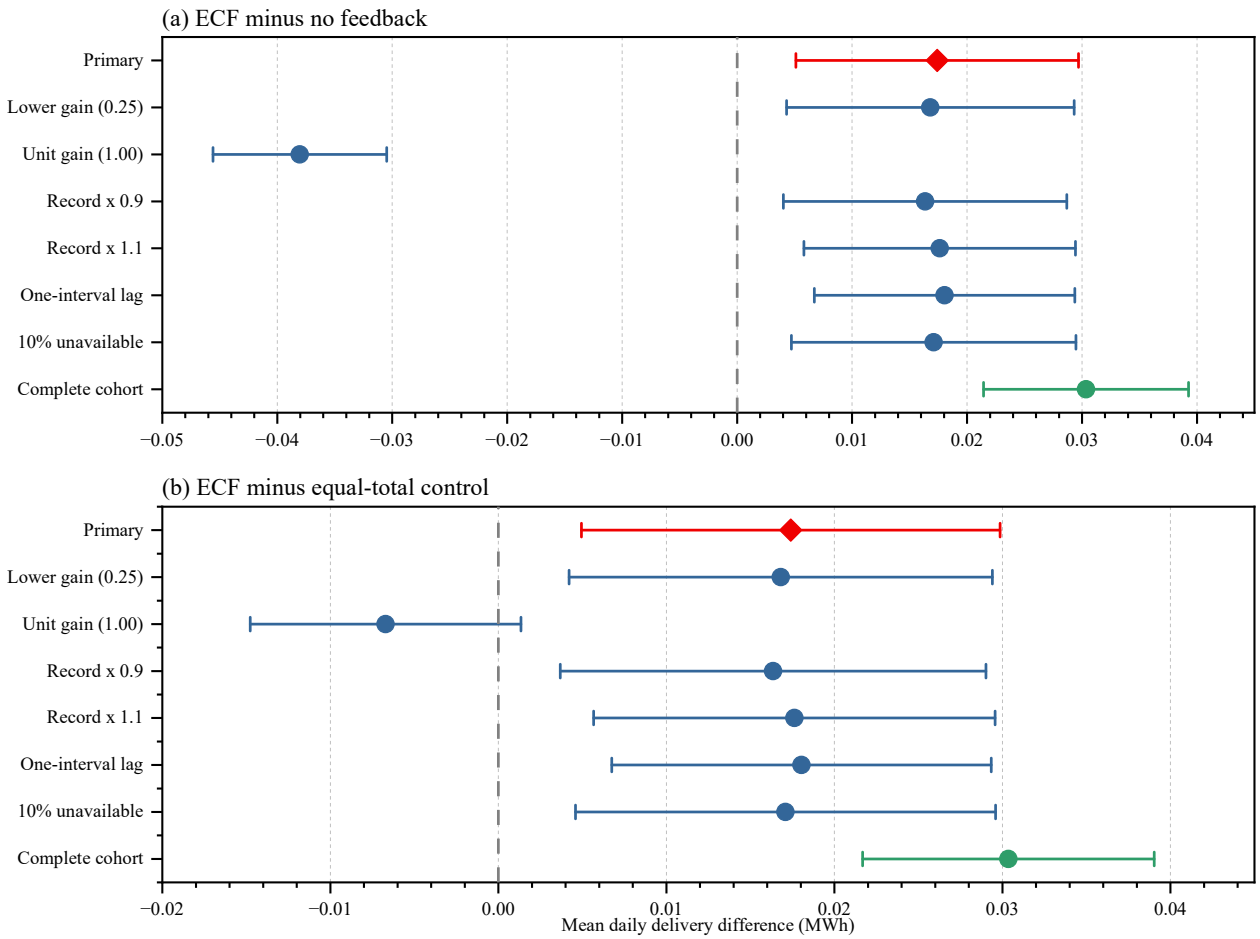


Figure 8: Sensitivity of the Australian replay to feedback gain, post-event record, and customer cohort. Markers show paired mean delivery differences per network day. Bars show 95% intervals adjusted for short-range dependence between neighboring dates. (a) ECF minus no feedback. (b) ECF minus the equal-total control. The red diamond denotes the primary unperturbed-proxy result, and blue circles denote gain or record changes. Green markers denote the complete-customer cohort, which uses the same load multiplier but changes customer composition and calendar coverage. Every transformed available-export record is floored at modeled delivery. All settings pass the specified allocation and delivery checks; the zero line separates positive from negative mean delivery differences.

MW between the smallest and largest scenarios. More PV does not imply a proportionally larger feedback benefit (Supplementary Section S13.2).

Changing the network section gives a related spatial comparison. ECF delivers more than the equal-total control in all 13 published LV sections. The 80-day differences range from 71.2 to 1,311.6 kWh. Relative to no feedback, 12 sections gain delivery; LV26 loses 3.7 kWh. The section trajectories pass the stated physical checks. These related models come from one parent collection, and their separate gains are not summed into a feeder estimate (Supplementary Section S13.3).

The representative-topology panel also contains null responses. Six of the nine representative LV models contain qualifying replay intervals and show positive aggregate delivery differences against both controls. All four control trajectories pass the specified physical checks in every model. J and

Q show the largest gains, whereas L, N and O have no qualifying interval and remain unchanged (Supplementary Fig. 5). These published networks and their assigned operating conditions produce different responses[37].

Spatially uneven gains also appear within the primary feeder. Of its 418 participants, 330 deliver more over the 80 days than without feedback and 88 deliver less; the largest individual loss is 7.3 kWh. An aggregate gain does not imply that every connection benefits (Supplementary Section S14.1).

3 Discussion

An unused export award has no fixed cost to the rest of the feeder. The controlled comparisons find both productive large requests and requests that protect one connection's delivery while reducing delivery elsewhere. A shortfall charge based on one meter cannot distinguish them.

Spreading the same request reduction across all participants does not reproduce targeted feedback. Updating after every unused award is also less productive than checking for recoverable delivery. Its much higher update frequency is not the whole explanation: a replay-free update selected at a similar frequency delivers less too. The unqualified update leaves less authorization unused but removes useful access. Neither the change in total authorization nor unused authorization alone determines delivery.

A successful comparison for one completed interval does not guarantee that a tight ceiling will help in the next. Available power can recover. The loss at unit feedback gain, and its persistence when ceiling memory continues across observed date blocks, show that the size of the later update matters as well as the reason for making it. Record error creates another tradeoff: a conservative availability bound prevents unsupported replay decisions in the tested cases, but can miss opportunities and lower subsequent delivery. A deployment would need to assess delivered energy and ceiling recovery over time, together with the accuracy and cost of obtaining the post-event record.

The public-data simulations use measured household chronology and several published network representations. The primary feeder's added delivery is largely reflected in reduced source-terminal import. The gain remains positive across the tested profile placements and PV capacities, but its size changes; one continuous date block also contributes most of the primary gain. Some network models have no qualifying intervals, and 88 of the 418 primary-feeder participants deliver less than under no feedback. The replay qualifies an unused-award group; it does not assign individual causal responsibility within that group.

The controlled study verifies allocation and delivery against voltage, balance, and branch apparent-power limits at both ends with declared synthetic branch ratings. The public study checks customer voltages and native transformer limits, using constructed requests and a profile-derived availability record rather than linked operating requests and telemetry. The replay is meaningful when its availability estimate remains valid under the alternative award. Storage dispatch, flexible demand, or inverter control that responds to access would have to be represented in that comparison. Longer field records could test these information requirements and participant responses to future ceilings. Annual energy, economic value, and emissions require additional data and models.

Completed records can therefore guide a later request update when they establish a feasible opportunity for delivery elsewhere, without replacing the network limits or allocation rule.

4 Methods

4.1 Requests, awards and available export

Let $\mathcal{N}$ be the participating export connections and $\mathcal{G} = (\mathcal{B}, \mathcal{L})$ the distribution network. In interval t, participant i requests $r_i(t)$ before its available export $a_i(t)$ is known. Its fixed connection bound is C_i. Feedback changes a request ceiling $\psi_i(t) \in [0, C_i]$, initially C_i, so the admitted request is

$$r_i^{\mathrm{adm}}(t) = \min\{r_i(t), C_i, \psi_i(t)\}. \tag{1}$$

Allocation rule $\mathcal{R}$ then grants access $\mathbf{x}(t) = \Pi_{\mathcal{R}}(\mathbf{r}^{\mathrm{adm}}(t); \mathcal{F}_{\mathcal{M}}(\mathcal{G}, \mathbf{z}(t)), \boldsymbol{\eta}_{\mathcal{R}}(t))$. Here $\mathcal{F}_{\mathcal{M}}$ represents network equations and limits under exogenous data $\mathbf{z}(t)$; $\boldsymbol{\eta}_{\mathcal{R}}$ contains rule parameters. Completed delivery is $y_i(t) = \min\{x_i(t), a_i(t)\}$, so $x_i - y_i$ is the unused award. Feedback modifies only the admitted-request input in a later interval, not the rule or network limits. These variables are interval-average powers; their sums over half-hours give the reported energy totals. Supplementary Table 1 collects the corresponding quantities.

The coordinator does not observe participants' private output signals or conditional distributions before allocation. It observes completed delivery and uses a post-event available-export record $a_i^{\mathrm{tel}}(t)$ for the replay. The controlled experiment supplies exact simulated availability. The Australian simulation constructs a profile-derived proxy and uses it as the primary replay record; neither this proxy nor modeled delivery is a field measurement of curtailed export.

Assumption 1 (Post-event available-export information)**.** *The comparison uses an available-export record a_i^{tel} for the completed interval. Its origin and time alignment must support the delivery calculation under an alternative award. In the controlled study the record is exact simulated availability; in the Australian study the primary record is the profile-derived proxy a_i^{prof}. Record sensitivities alter that proxy only for the replay decision. Neither record changes an award already issued.*

Assumption 2 (Same-state replay)**.** *Only admitted requests change in the completed-interval comparison. Topology, exogenous network data, participants, rule parameters and underlying available export remain fixed. Awards and power flows are recalculated. If a different award would change availability through inverter controls, storage, demand or portfolio dispatch, those responses must instead enter the replay model.*

4.2 Request choice and delivery displacement

The controlled study links a participant's request to later delivery through one conditional distribution of available export. Participant i observes a private signal s_i, forms a point forecast a_i^{fcst}, and assigns conditional cumulative distribution $F_i(\cdot \mid s_i)$ to its available export. Let $r_i^{\mathrm{ref}} = a_i^{\mathrm{fcst}}$ be its point-forecast request and $r_i^{\mathrm{chg}} = \gamma_i r_i^{\mathrm{ref}}$, $\gamma_i \geq 1$, a larger request. Holding other requests and the network state fixed, the participant values own delivery and faces a modeled shortfall cost $\vartheta \geq 0$ per unit of unused award, normalized by the value of delivered export:

$$U_i(\gamma_i; \vartheta) = \mathbb{E}\big[\min\{x_i(\gamma_i), a_i\} - \vartheta[x_i(\gamma_i) - a_i]^+ \mid s_i\big]. \tag{2}$$

The selected multiplier is the smallest maximizer on the prespecified grid Γ. The same conditional available-export distribution determines this choice and the subsequent realization; the request and delivery curves are not generated independently. A static changed-request case is classified as a request–access–delivery externality if the changed group's expected payoff and unused award both

rise while aggregate expected delivery falls beyond the numerical tolerance. A productive high request raises the group's payoff without such a delivery loss. Dynamic cases use realized counterparts. This economic classification is distinct from ECF's operational replay test.

Within a smooth allocation regime, write $J_{ji} = \partial x_j / \partial r_i$ and $\bar{Y} = \sum_j \mathbb{E}[\min\{x_j, a_j\} \mid s_j]$ for expected total delivery. Here $F_j(x_j)$ abbreviates the conditional cumulative distribution $F_j(x_j \mid s_j)$. Then

$$\begin{aligned} \frac{\partial U_i}{\partial r_i} &= J_{ii}[1 - (1+\vartheta)F_i(x_i)], \\ \frac{\partial \bar{Y}}{\partial r_i} &= \sum_j J_{ji}[1 - F_j(x_j)]. \end{aligned} \tag{3}$$

The first derivative need not have the sign of the second because the private payoff excludes delivery lost elsewhere. Where a request change crosses a bound or active network constraint, all reported classifications instead use matched finite differences. When both quantities are positive, outside-delivery loss divided by the additional unused award gives the diagnostic ratio plotted in Fig. 3(d). Its finite-change definition and relation to the local marginal shortfall cost are in Supplementary Section S2; the capacity-normalized Jain index is defined in Section S8.

4.3 Allocation rules and physical checks

Five rules use the same admitted requests and controlled-study network domain (Supplementary Table 2). Weighted proportional (WP) scales all requests by one feasible factor. Equal reduction (ER) subtracts a common amount; equal access (EA) applies a common cap. Maximum export (ME) maximizes total authorized access. Fairness-aware allocation (FA) trades authorized access against a request-normalized equality term. Their one-parameter formulas, conic objectives, deterministic tie-breaks and sensitivity derivations are specified in Supplementary Sections S4–S8. All rules allocate against an affine second-order-cone approximation of voltage and two-ended branch limits; the approximation is not treated as the final AC feasibility check.

The controlled medium-voltage case uses the published radial 141-bus network [38], a 0.70 demand scale and 30 fixed connection points. Synthetic, frozen branch ratings define this mechanism experiment; they are not native equipment nameplates. Q0 denotes zero participant reactive power and Q95 denotes 0.95 power factor with reactive-power absorption. Each matched reference, changed-request, award, observed-delivery and replay-delivery state is checked by nonlinear radial AC power flow at its fixed reactive-power setting. The check requires convergence, active- and reactive-power balance, $0.95 \le |V_b| \le 1.05$ pu and

$$\frac{\max_{e \in \{\mathrm{fr},\mathrm{to}\}} |S_{\ell,e}|}{\bar{S}^{\mathrm{syn}}_{\ell}} \le 1 \quad \text{for every monitored branch } \ell. \tag{4}$$

The branch test uses MVA at both ends, independently of rooted flow orientation. Supplementary Section S9 gives the frozen rating policy, calibration, and AC verification details.

4.4 Completed-interval replay and feedback

With unused-award tolerance $\epsilon_x > 0$, let $\mathcal{N}_{\mathrm{ua}}(t) = \{i : x_i(t) - y_i(t) > \epsilon_x\}$. After completion, ECF replaces only this group's admitted requests by the smaller of admission and metered delivery:

$$r_i^{\mathrm{rpl}}(t) = \begin{cases} \min\{r_i^{\mathrm{adm}}(t), y_i(t)\}, & i \in \mathcal{N}_{\mathrm{ua}}(t), \\ r_i^{\mathrm{adm}}(t), & i \notin \mathcal{N}_{\mathrm{ua}}(t). \end{cases} \tag{5}$$

The same allocation rule and network data give replay access $\mathbf{x}^{\mathrm{rpl}}$. Available export then bounds replay delivery:

$$y_i^{\mathrm{rpl}} = \min\{x_i^{\mathrm{rpl}}, a_i^{\mathrm{tel}}\}.$$

The completed award and delivery are never overridden. The test qualifies the episode only if the replay releases access outside the unused-award group, raises delivery for at least one outside participant, raises aggregate delivery, and passes the applicable nonlinear AC checks. Each positive change must exceed τ_{rpl}. The test qualifies a set-level opportunity; it does not attribute a loss to each member individually. With a validated record-error bound $|a_i^{\mathrm{tel}} - a_i| \le \delta_i$, the conservative replay uses

$$\max\{y_i, a_i^{\mathrm{tel}} - \delta_i\}$$

as a lower bound on available export. Supplementary Section S3 proves the resulting lower bound on recovered delivery. Physical checks remain separate. With exact availability and the same recovery threshold, positive aggregate delivery recovery implies positive outside delivery and access recovery. We retain all three quantities to show how recovery occurs and to evaluate perturbed records; they are not three independent filters in the exact-record model.

Let $g(t) = 1$ when all replay conditions hold and zero otherwise. The target and next-interval ceiling are

$$\psi_i^{\mathrm{tar}}(t) = \begin{cases} y_i(t), & g(t) = 1 \text{ and } i \in \mathcal{N}_{\mathrm{ua}}(t), \\ C_i, & \text{otherwise}, \end{cases}$$

$$\psi_i(t+1) = (1 - \beta_{\mathrm{ECF}})\psi_i(t) + \beta_{\mathrm{ECF}}\psi_i^{\mathrm{tar}}(t), \qquad 0 \le \beta_{\mathrm{ECF}} \le 1. \tag{6}$$

Thus only a qualifying episode draws its unused-award group's ceilings toward delivery. Otherwise ceilings recover toward connection bounds. Consecutive nonqualifying intervals give geometric recovery for positive gain; a larger raw request cannot bypass a binding ceiling.

The shortfall-cost benchmark changes request choice but has no feedback state. Mismatch-only feedback updates a smoothed delivery-to-award ratio without replay. A second ablation applies ECF's update after every unused award without qualification. To assess whether update frequency alone explains ECF's advantage, a diagnostic control selects unused-award intervals at a fixed probability. The probability is 483/3091, the number of ECF qualifications divided by unused-award intervals in the original Australian evaluation. A hash of a fixed seed, date and half-hour selects intervals without consulting the replay or availability record beyond the completed delivery needed to determine the unused-award group. We report all three fixed schedules and average them within each date. This comparison was specified after the original results were known; it is a diagnostic matched in expected update frequency, not an independent validation. Every schedule follows its own awards, delivery and ceiling state.

The equal-total control matches ECF's total request reduction in each interval but spreads it proportionally across connections. If $r_i^{\mathrm{cap}} = \min\{r_i, C_i\}$ and $\Delta R_{\mathrm{ECF}} = \sum_i (r_i^{\mathrm{cap}} - r_i^{\mathrm{adm,ECF}})$, the control admits

$$r_i^{\mathrm{adm,uni}} = r_i^{\mathrm{cap}} \left(1 - \frac{\Delta R_{\mathrm{ECF}}}{\sum_j r_j^{\mathrm{cap}}}\right) \tag{7}$$

when the denominator is positive, and retains the capacity-limited request otherwise. This isolates where requests are reduced. With total authorization $X = \sum_i x_i$ and unused authorization $H = \sum_i (x_i - y_i)$, the accounting identity $Y = X - H$ yields

$$Y^{\mathrm{ECF}} - Y^{\mathrm{uni}} = (X^{\mathrm{ECF}} - X^{\mathrm{uni}}) + (H^{\mathrm{uni}} - H^{\mathrm{ECF}}). \tag{8}$$

It does not assert that either term has a fixed sign. Supplementary Section S4 derives the corresponding redistribution conditions.

4.5 Controlled experiments

A five-node conditional available-export tree uses standardized support $(-2, -1, 0, 1, 2)$ with probabilities $(1, 4, 6, 4, 1)/16$ and scale σ. The participant's forecast anchors both its request and the later node realization. The static grid spans 30 locations, five rules, Q0 and Q95, five uncertainty levels and nine shortfall-cost levels, giving 13,500 request-choice cases. The selected payoff-maximizing multiplier is held fixed within a 16-interval trajectory. Three trajectories have identical node counts but differ in ordering: clustered, intermittent and shock–recovery. Changed requests come from one participant, five electrically nearby participants, or five dispersed participants. In the five-participant cases, each uses its own best-response multiplier and all five requests change simultaneously. The matched design comprises 1,620 trajectories and 25,920 interval comparisons across the controls and gains. Supplementary Table 3 gives every factor, numerical tolerance and parameter level. The post-event record is exact simulated availability in this controlled study.

4.6 Australian public-data simulation

The external simulation combines a published unbalanced 11-kV feeder with 1,255 service connections and its LV sections[39] with half-hour PV and consumption profiles from 300 Ausgrid solar homes[40]. These sources have no common customer identifier. The primary deterministic profile-to-service mapping is fixed before assessment, and separate placements test its sensitivity. The forward-uptake scenario assigns 418 connections export participation and scales PV to an approximately 9.9-kWp mean[41]. It is not a historical reconstruction of issued operating envelopes. The day-to-day forecast increment is estimated from 2010–2011 data; 2011–2012 data select fixed feeder settings; 2012–2013 provide the assessment chronology. A day is eligible only when mapped profiles are complete on that day and the previous day. Excluding missing and daylight-saving-transition records leaves 80 network days, or 3,840 half-hours per control.

For mapped profile $\iota(i)$, scaled gross PV generation GG, general consumption GC and controlled load CL yield

$$a_i^{\mathrm{prof}}(d,\tau) = \left[\lambda_{\mathrm{PV}}\mathrm{GG}_{\iota(i)}(d,\tau) - \lambda_L(\mathrm{GC}_{\iota(i)}(d,\tau) + \mathrm{CL}_{\iota(i)}(d,\tau))\right]^+ . \tag{9}$$

Modeled delivery under control c is $y_i^c = \min\{x_i^c, a_i^{\mathrm{prof}}\}$; the unperturbed profile proxy also supplies the primary replay record. Raw requests use the previous day's same-half-hour proxy plus a positive 90th-percentile increment from same-half-hour day-to-day changes. Contemporaneous household demand is fixed as an exogenous input to this retrospective allocation; load-forecast error is not modeled.

The primary feeder settings use a 0.30 load multiplier and a 1.07-pu allocation-stage upper-voltage limit, below the 1.10-pu physical limit. For each half-hour, a zero-anchored feasible-factor search scales admitted requests under nonlinear unbalanced AC power flow. An independent factor-grid check of all 19,200 primary allocation calls found no feasible sampled factor above the returned one. The sampled check supports these allocations but does not prove global maximality between grid points (Supplementary Section S10.4). Load-only, authorized-export and modeled-delivery states are screened separately for convergence, customer voltage and native transformer ratings. Because the public model lacks usable native line ampacities, line-thermal certification is outside this physical

screen. The replay threshold is 2 W, a numerical decision threshold rather than an instrument-accuracy claim.

Each primary day begins with ceilings at C_i and compares no feedback, previous-day persistence, ECF at $\beta_{\mathrm{ECF}} = 0.5$, and equal-total reduction. The additional comparisons vary gain, replay qualification, cross-day memory, record perturbations and error bounds, profile placement, PV capacity, participant adaptation and published LV networks while retaining the documented fixed inputs for each matched comparison. Source-terminal power from the same AC solves accounts for upstream exchange. The request quantile and fixed feeder settings are calculated from the earlier source years. Earlier evaluations had already been inspected when the present scenario was constructed, so its 80-day comparison is a fixed-scenario reevaluation rather than a new blind validation. Supplementary Section S10.3 records the selection sequence and full model construction.

4.7 Statistical analysis

The primary comparison unit in the Australian study is one paired network day. A day contains 48 half-hours on the same network and profile mapping; participants, half-hours and nested LV sections are not treated as independent network samples. Energy differences are computed within each day before averaging. We report ordinary paired-day percentile intervals and a date-dependent wild-bootstrap interval that allows short-range correlation between nearby eligible dates. Both use 10,000 replicates. In the request adaptation and frequency-calibrated-update experiments, three random seeds are averaged within each date before their comparisons enter this calculation. Intervals are pointwise and do not adjust for multiple comparisons. Supplementary Section S10.3 gives the calendar covariance, exact eligibility rules and full numerical settings.

Author contributions

Junjie Huang: conceptualization, methodology, software, formal analysis, investigation, visualization, data curation, and writing—original draft. Tao Yu and Yusheng Xue: conceptualization, supervision, funding acquisition, and writing—review and editing. Zhenning Pan, Yufeng Wu, and Wencong Xiao: methodology, validation, and writing—review and editing. Rankai Zhu: validation, data curation, and writing—review and editing. Jie Huang: validation and writing—review and editing.

Competing interests

The authors declare no competing interests.

Data availability

Calculated results, figure source data, input-source identifiers, preprocessing instructions, date selections and network-profile mapping tables are available in the public project repository, `https://github.com/MarxHuang/ecf-export-access`. The source 141-bus network, CSIRO/GridQube network collection (doi:10.25919/ghnz-bk28) and Ausgrid Solar Home Electricity Data remain with their original distributors under their respective terms; raw third-party records are not redistributed.

The repository distinguishes deposited calculated data from those external inputs required for a full simulation rerun.

Code availability

Simulation, preprocessing and analysis code, parameter files, environment specifications and reproduction instructions are available at `https://github.com/MarxHuang/ecf-export-access`. The repository identifies its deposited revision and provides checks of saved results separately from complete reruns, which require the source datasets described above.

Acknowledgements

This work was funded by State Grid Electric Power Research Institute Co., Ltd., Project "Interaction and Coordination Technology of Cyber-Physical-Social Elements (II)" (SGNR0000KJJS2505736).

References

[1] K. Petrou, A. T. Procopiou, L. Gutiérrez-Lagos, M. Z. Liu, L. F. Ochoa, T. Langstaff, and J. M. Theunissen. Ensuring distribution network integrity using dynamic operating limits for prosumers. *IEEE Trans. Smart Grid*, 12(5):3877–3888, 2021. doi: 10.1109/TSG.2021.3081371.

[2] M. Z. Liu, L. F. Ochoa, P. K. C. Wong, and J. M. Theunissen. Using OPF-based operating envelopes to facilitate residential DER services. *IEEE Trans. Smart Grid*, 13(6):4494–4504, 2022. doi: 10.1109/TSG.2022.3188927.

[3] M. M. Hoque, M. Khorasany, M. I. Azim, R. Razzaghi, and M. Jalili. Dynamic operating envelope-based local energy market for prosumers with electric vehicles. *IEEE Trans. Smart Grid*, 15(2):1712–1724, 2024. doi: 10.1109/TSG.2023.3302270.

[4] SA Power Networks. Distribution annual planning report 2024/25 to 2028/29. Technical report, SA Power Networks, December 2024. URL `https://www.sapowernetworks.com.au/public/download.jsp?id=9716`. accessed 3 September 2026.

[5] California Public Utilities Commission. Resolution e-5230: Proposals to implement limited generation profiles and time-varying inverter export values. Technical report, California Public Utilities Commission, December 2022. URL `https://docs.cpuc.ca.gov/PublishedDocs/Published/G000/M499/K779/499779492.PDF`. issued 9 December 2022; accessed 3 September 2026.

[6] European Parliament and Council of the European Union. Directive (EU) 2024/1711 amending directives (EU) 2018/2001 and (EU) 2019/944 as regards improving the union's electricity market design. Official Journal of the European Union, 2024. URL `https://eur-lex.europa.eu/eli/dir/2024/1711/oj`. accessed 3 September 2026.

[7] National Energy Administration of China. Policy interpretation of the measures for the development and construction of distributed photovoltaic power generation, January 2025. URL `https://www.nea.gov.cn/20250123/d38e5436b4d04159863ddbc10a6ede10/c.html`. in Chinese; accessed 3 September 2026.

[8] M. Mallaki, M. S. Naderi, M. Abedi, S. D. Manshadi, and G. B. Gharehpetian. Strategic bidding in distribution network electricity market focusing on competition modeling and uncertainties. *J. Mod. Power Syst. Clean Energy*, 9(3):561–572, 2021. doi: 10.35833/MPCE.2019.000177.

[9] D. Ellman and Y. Xiao. Incentives to manipulate demand response baselines with uncertain event schedules. *IEEE Trans. Smart Grid*, 12(2):1358–1369, 2021. doi: 10.1109/TSG.2020.3024208.

[10] X. Wang and W. Tang. Modeling and analysis of baseline manipulation in demand response programs. *IEEE Trans. Smart Grid*, 13(2):1178–1186, 2022. doi: 10.1109/TSG.2021.3137098.

[11] X. Yu and Y. Xue. Smart grids: A cyber–physical systems perspective. *Proc. IEEE*, 104(5):1058–1070, May 2016. doi: 10.1109/JPROC.2015.2503119.

[12] Y. Xue and X. Yu. Beyond smart grid—cyber–physical–social system in energy future [point of view]. *Proc. IEEE*, 105(12):2290–2292, December 2017. doi: 10.1109/JPROC.2017.2768698.

[13] B. Liu and J. H. Braslavsky. Robust dynamic operating envelopes for DER integration in unbalanced distribution networks. *IEEE Trans. Power Syst.*, 39(2):3921–3936, 2024. doi: 10.1109/TPWRS.2023.3308104.

[14] B. Liu, J. H. Braslavsky, and N. Mahdavi. Linear OPF-based robust dynamic operating envelopes with uncertainties in unbalanced distribution networks. *J. Mod. Power Syst. Clean Energy*, 12(4):1320–1326, 2024. doi: 10.35833/MPCE.2023.000653.

[15] Australian Energy Regulator. Export limits guidance note, 2024. URL `https://www.aer.gov.au/industry/registers/resources/guidelines/export-limit-guidance-note/final`. accessed 13 August 2026.

[16] J. Iria, P. Scott, D. Gordon, A. Attarha, and S. M. Noori R. A. Trial of shaped operating envelopes: Final technical knowledge sharing report. Technical report, The Australian National University, March 2024. URL `https://arena.gov.au/knowledge-bank/project-converge-final-technical-report/`. accessed 13 August 2026.

[17] Australian Energy Market Operator. Project edge data specification: Part a—on-boarding and enrolment data. Technical report, Australian Energy Market Operator, 2023. URL `https://www.aemo.com.au/-/media/files/initiatives/der/2023/project-edge---data-specification-part-a.pdf`. accessed 13 August 2026.

[18] K. Petrou, M. Z. Liu, A. T. Procopiou, L. F. Ochoa, J. M. Theunissen, and J. Harding. Operating envelopes for prosumers in LV networks: A weighted proportional fairness approach. In *Proc. IEEE PES Innov. Smart Grid Technol. Europe (ISGT-Europe)*, pages 579–583, 2020. doi: 10.1109/ISGT-Europe47291.2020.9248975.

[19] M. R. Alam, P. T. H. Nguyen, L. Naranpanawe, T. K. Saha, and G. Lankeshwara. Allocation of dynamic operating envelopes in distribution networks: Technical and equitable perspectives. *IEEE Trans. Sustain. Energy*, 15(1):173–186, 2024. doi: 10.1109/TSTE.2023.3275082.

[20] J. J. Cuenca and B. P. Hayes. Non-bias allocation of export capacity for distribution network planning with high distributed energy resource integration. *IEEE Trans. Power Syst.*, 37(4):3026–3035, 2022. doi: 10.1109/TPWRS.2021.3124999.

[21] Z. Jiang and Y. Guo. Bargaining-based approach for dynamic operating envelope allocation in distribution networks. *IEEE Trans. Smart Grid*, 16(5):3588–3600, 2025. doi: 10.1109/TSG.2025.3566419.

[22] M. Z. Liu, A. T. Procopiou, K. Petrou, L. F. Ochoa, T. Langstaff, J. Harding, and J. M. Theunissen. On the fairness of PV curtailment schemes in residential distribution networks. *IEEE Trans. Smart Grid*, 11(5):4502–4512, 2020. doi: 10.1109/TSG.2020.2983771.

[23] P. Lusis, L. L. H. Andrew, S. Chakraborty, A. Liebman, and G. Tack. Reducing the unfairness of coordinated inverter dispatch in PV-rich distribution networks. In *Proc. IEEE Milan PowerTech*, pages 1–6, 2019. doi: 10.1109/PTC.2019.8810940.

[24] R. K. Gupta and D. K. Molzahn. Analysis of fairness-promoting optimization schemes of photovoltaic curtailments for voltage regulation in power distribution networks. In *Proc. IEEE Kiel PowerTech*, pages 1–7, 2025. doi: 10.1109/PowerTech59965.2025.11180583.

[25] R. K. Gupta, P. Buason, and D. K. Molzahn. Fairness-aware photovoltaic generation limits for voltage regulation in power distribution networks using conservative linear approximations. In *Proc. IEEE Texas Power Energy Conf. (TPEC)*, pages 1–6, 2024. doi: 10.1109/TPEC60005.2024.10472277.

[26] M. Vassallo, A. Bolland, A. Bahmanyar, L. Wehenkel, L. Duchesne, D. Liu, S. Khaskheli, A. H. Thuc, P. P. Vergara, A. Anvari-Moghaddam, S. Gerard, and D. Ernst. SecuLEx: A secure limit exchange market for dynamic operating envelopes. *Electr. Power Syst. Res.*, 263:113736, 2027. doi: 10.1016/j.epsr.2026.113736.

[27] M. U. Hashmi and D. Van Hertem. Robust dynamic operating envelopes for flexibility operation using only local voltage measurement. In *Proc. Int. Conf. Smart Energy Syst. Technol. (SEST)*, pages 1–6, 2023. doi: 10.1109/SEST57387.2023.10257504.

[28] H. Moring, X. Farrell, and J. L. Mathieu. Fair-over-time distributed energy resource coordination. In *Proc. 60th Annu. Allerton Conf. Commun., Control, Comput.*, pages 1–8, 2024. doi: 10.1109/Allerton63246.2024.10735316.

[29] S. Sweeney, P. Kilby, B. Penney, K. Moghaddasi, and S. Mudiyanselage. Stateful pricing and allocation for repeated constrained DER coordination in distribution networks. arXiv:2606.22463, 2026. Preprint, version 1.

[30] Y. Chen, C. Zhao, S. H. Low, and A. Wierman. An energy sharing mechanism considering network constraints and market power limitation. *IEEE Trans. Smart Grid*, 14(2):1027–1041, 2023. doi: 10.1109/TSG.2022.3198721.

[31] G. Tsaousoglou, J. S. Giraldo, P. Pinson, and N. G. Paterakis. Mechanism design for fair and efficient DSO flexibility markets. *IEEE Trans. Smart Grid*, 12(3):2249–2260, 2021. doi: 10.1109/TSG.2020.3048738.

[32] G. Belgioioso, W. Ananduta, S. Grammatico, and C. Ocampo-Martinez. Operationally-safe peer-to-peer energy trading in distribution grids: A game-theoretic market-clearing mechanism. *IEEE Trans. Smart Grid*, 13(4):2897–2907, 2022. doi: 10.1109/TSG.2022.3158442.

[33] G. Tsaousoglou, K. Mitropoulou, K. Steriotis, N. G. Paterakis, P. Pinson, and E. Varvarigos. Managing distributed flexibility under uncertainty by combining deep learning with duality. *IEEE Trans. Sustain. Energy*, 12(4):2195–2204, October 2021. doi: 10.1109/TSTE.2021.3086846.

[34] C. Essayeh, I. Savelli, and T. Morstyn. Optimising flexibility procurement in local flexibility markets under stochastic availability of participants. *Sustain. Energy Grids Netw.*, 44:102061, December 2025. doi: 10.1016/j.segan.2025.102061.

[35] K. L. Anaya and M. G. Pollitt. How to procure flexibility services within the electricity distribution system: Lessons from an international review of innovation projects. *Energies*, 14(15): 4475, 2021. doi: 10.3390/en14154475.

[36] X. Wang and W. Tang. A self-reported baseline demand response program for mitigation of baseline manipulation. In *Proc. IEEE Power Energy Soc. Gen. Meeting (PESGM)*, pages 1–5, 2021. doi: 10.1109/PESGM46819.2021.9638109.

[37] F. Geth, T. Brinsmead, S. West, P. Goldthorpe, B. Spak, G. Cross, and J. Braslavsky. National low-voltage feeder taxonomy study. Technical report, CSIRO, 2021. URL `https://doi.org/10.25919/2tas-7213`.

[38] H. M. Khodr, F. G. Olsina, P. M. De Oliveira-De Jesus, and J. M. Yusta. Maximum savings approach for location and sizing of capacitors in distribution systems. *Electr. Power Syst. Res.*, 78(7):1192–1203, 2008. doi: 10.1016/j.epsr.2007.10.002.

[39] F. Geth, R. Heidari, J. Clark, K. Lucas, and N. Nimalsiri. Realistic australian medium voltage feeder with associated low voltage feeders, 2025.

[40] Ausgrid. Solar home electricity data, 1 july 2010 to 30 june 2013, 2014. URL `https://data.gov.au/data/dataset/nsw-solar-home-electricty-data`. 300 gross-metered homes; accessed 13 August 2026.

[41] Clean Energy Regulator. Quarterly carbon market report: March quarter 2025. Technical report, Australian Government Clean Energy Regulator, 2025. URL `https://cer.gov.au/document/quarterly-carbon-market-report-march-quarter-2025`. accessed 13 August 2026.